Sonali Sharma

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SUMMARY

Geospatial Analyst and Environmental Scientist with 7+ years of experience delivering spatial analysis and GIS workflows across research and interdisciplinary projects. Strong expertise in 2D–3D remote sensing datasets, GIS modelling, and scalable geospatial workflow design using ArcGIS, QGIS, Python and R. Skilled at translating complex environmental datasets into actionable spatial insights that support evidence-based planning and real-world decision-making. Proven track record of delivering results across international, interdisciplinary & cross-functional teams. Motivated to apply blend of scientific foundation and hands-on GIS expertise within applied consultancy, urban, nature & environmental planning contexts.

SKILLS

GIS & Statistical analyses; Visualization: QGIS, ArcGIS Pro, GeoServer, Google Earth Engine, Spatial Databases
Programming & Automation : Python, R, JavaScript, HPC workflows, UNIX/Linux
Project Management: Strategic planning, Interpretation and Decision-making
Communication and Collaboration: Cross-functional & Cross-cultural Teamwork, Adaptability, Clear communication, Confident demeanour
Research: Scientific writing, Experimental design, Analytical thinking

LANGUAGES

English (C2 - Fluent), German (A2; eingeschrieben in den B1-Kurs)

WORK EXPERIENCES

Postdoctoral Researcher
University of Helsinki, Finland Sep 2023–Oct 2025

- Designed and automated GIS workflows (Python, R, HPC) integrating LiDAR and high-resolution land cover data, reducing processing time by 70%.
- Developed a reproducible machine learning-based pipeline for 3-D urban spatial analysis.
- Coordinated interdisciplinary GIS project with Finnish Environment Institute, delivering actionable map products and scientific outputs.
- Configured and managed spatial datasets to support biodiversity and urban planning projects.
- Presented findings at interdisciplinary workshops with municipal authorities, workshops and conference.

Visiting Doctoral Researcher
Martin-Luther-University (Germany) Aug 2020–Sept 2021

- Being DAAD Bi-national Fellowship recipient collaborated with German research teams
- Led research on future plausible urban growth and climate change scenarios (ArcGIS, Python, R, UNIX/LINUX, & GEE)
- Presented and participated in international conferences and workshops focused on urban sustainability and geospatial methods

Graduate Research Fellow (PhD)
Jawaharlal Nehru University (India) Aug 2018–Jun 2023

- Modeled urban ecosystem services: soil erosion, flood regulation, local heat regulation and carbon sequestration; using ArcGIS, QGIS, Python, R, and Google Earth Engine.
- Integrated multi-source datasets (climate, land cover, remote sensing) for urban growth and heat island studies.
- Delivered outputs including geospatial models, reports, and spatial analysis frameworks for planning.
- Supported and mentored junior researchers in GIS workflows and data processing.
- Presented and participated in international conferences, seminars, workshops, and summer schools focused on sustainability and geospatial methods

CERTIFICATIONS

Fundamentals of GeoAI: Deep Learning for Geospatial Analysis
Instructor: Milan Janosov (Udemy) 2026

- Gained a theoretical and practical understanding to build and train U-Net model for urban image segmentation and creating their visual interactive outputs.

Urban Analytics in Python
Instructor: Milan Janosov (Udemy) 2026

- Gained hand-on functional knowledge of Python for Urban livability analytics.

Location & AI – Basic & Deep Dive into Practical Solutions
Instructor: Location Innovation Hub 2025

- Gained a holistic understanding of GeoAI technologies, enhancing skills relevant to AI/ML, GIS, environmental analysis, and urban planning.

Spatial Machine Learning and Statistics in Python
Instructor: Milan Janosov (LinkedIN) 2025

- Got a hands-on introduction to the intersection of machine learning and spatial analytics

Going Places with Spatial Analysis and Make an Impact with Modern Geo Apps
ESRI MOOCs 2024–2025

- Developed hands-on expertise in cloud-based spatial analysis and built interactive geo apps and dashboards to support informed decision-making

Pedagogy 1 - Learning in higher education
University of Helsinki 2024–2025

- Reflected on educator's role and pedagogical impact; explored inquiry-based approaches to foster motivation and deep learning

Urban Nature: Connecting Cities, Sustainability, and Innovation
Lund University: Coursera 2024

- Explored collaborative and governance strategies for implementing nature-based solutions in multifunctional ways, with practical case studies

EDUCATION

Master of Science in Environmental Sciences
Central University of Rajasthan, India 2015–2017

- Courses: Atmospheric Sciences, Ecology, Geography, Geology, Remote Sensing-GIS, and Oceanography

Bachelor of Science in Biology
Mohan Lal Sukhadia University, India 2012–2015

- Courses: Botany, Zoology, and Chemistry

PUBLICATIONS

Peer reviewed : 6

1. Dennis, M., Huck, J., Holt, C., McHenry, E., Andersson, E., **Sharma, S**., & Haase, D. (2026). In search of Schrödinger's patch: A functional approach to habitat delineation. Landscape Ecology, 41, 36.

2. Dennis, M., Huck, J., Holt, C., McHenry, E., Andersson, E., **Sharma, S**., & Haase, D. (2026). Beyond the patch: Leveraging the notion of realized habitat in fragmentation-biodiversity research. Landscape Ecology, 41, 37.

3. **Sharma, S**., Joshi, P.K., & Fürst, C. (2022). Exploring Multiscale Influence of Urban Growth on Landscape Patterns of Two Emerging Urban Centers in the Western Himalaya. Land, 11, 2281.

4. **Sharma, S**., Joshi, P.K., & Fürst, C. (2022). Unravelling net primary productivity dynamics under urbanization and climate change in the Western Himalaya. Ecological Indicators, 144, 109508.

5. **Sharma, S**., Sharma, M., Anees, M.M., & Joshi, P.K. (2020). Longitudinal study of changes in ecosystem services in a city of lakes, Bhopal, India. Energy, Ecology and Environment, 6, 408–424.

6. **Sharma, S**., Nahid, S., Sannigarhi, S., Sharma, M., Anees, M.M., Sharma, R., Shekhar, R., Joshi, P.K., Basu, A.S., Pilla, F., & Basu, B. (2020). A long-term and comprehensive assessment of urbanization-induced impacts on ecosystem services in the capital city of India. City and Environment Interactions, 7, 1-12, 100047.

Under revision/review: 2

1. **Sharma, S**., Maija, T., & Andersson, E. (n.d.). Situating Private Urban Green Spaces (PrUGSs) - using transfer learning to characterise urban landscapes (Urban Forestry and Urban Greening)

2. **Sharma, S**., Dennis, M., & Andersson, E. (n.d.). Qualifying and navigating urban heterogeneity for biodiversity. (Ecography).

To be communicated shortly: 1

1. **Sharma, S**., Joshi, P.K., & Fürst, C. (n.d.). Rampant or Ecological? Urban growth pathways in Himalaya. (Ecological Indicators).

INTERESTS

Traveling, Yoga, Meditation, Forest walks, Dance